Deviations from Covered Interest Rate Parity: The Case of British Pound Sterling versus Euro

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KEY FINDINGS

- We focus on the investigation of deviations from covered interest rate parity on the British pound and the euro and include event-driven factors.
- Arbitrage opportunities seem to be persistent and vary systematically. We make the driving factors explicit.
- The presence of nonlinearities requires the application of methods from deep learning. It is shown that deep learning adds value. Equipped with better forecasts, arbitrage desks can prepare for days when there are large arbitrage gains. Corporates can punctually adapt their procurement of currencies that are about to become scarce.

ABSTRACT

The authors find that the foreign exchange derivatives market for British pound sterling versus euro deviates from the covered interest rate parity (CIP). The resulting arbitrage opportunities seem to be persistent and vary systematically. They are driven not only by Brexit-related politics. The authors find a relation between the cross-currency basis and various factors. Furthermore, they discover nonlinearities that require the application of deep learning methods. The findings are important for arbitrage desks: They show when arbitrage opportunities will become large for international trade, when to look for better alternatives than hedging with forwards, and when corporate treasuries should procure currencies—that are about to become scarce—in advance.

TOPICS

*Big data/machine learning, currency, simulations**

Standard textbooks in international economics claim that "there is strong empirical evidence that the covered interest parity condition holds" and that "deviations occur … when asset holders fear that governments may impose regulations which prevent free movement of foreign funds" (Krugman and Obstfeld 2006). However, recent research paints a different picture: "We find that deviations from the covered interest rate parity (CIP) condition imply large, persistent, and systematic arbitrage opportunities in one of the largest asset markets in the world" (Du, Tepper, and Verdelhan 2018). These opportunities can be quantified by the cross-currency basis (CCB).

We go beyond the quoted study by deepening the investigation of the specific case of deviations from CIP in the market for the British pound (GBP) against the euro. Hence, the CCB is the difference between the direct euro interest rate from the cash market and the synthetic euro interest rate obtained by converting GBP into euro. A positive (negative) CCB means that the direct euro interest rate is higher (lower) than what one would get by swapping (Avdjiev et al. 2016). In other words, if the CCB is large, there is a flight into euros. "As is now well known, large bases appeared during the height of the global financial crisis and the European debt crisis, as the interbank-markets became impaired and arbitrage capital was limited" (Du, Tepper, and Verdelhan 2018). What exactly happened during the crisis? "USD funding markets became extremely expensive and at times completely inaccessible. Consequently, both U.S. banks and foreign banks with USD assets sought to borrow USD synthetically via EUR. … The sheer number of participants executing this trade, combined with limited capacity on the part of dealers, affected the price of forward contracts and pushed the forward exchange rate away from that implied by the CIP condition" (Pinnington and Shamloo 2016). After the global financial crisis, violations of CIP continue to exist within the European Union (EU) as well (Ferreira and Dionísio 2015), especially for the GBP (Ferreira and Kristoufek 2017). We show that a nonzero CCB for the GBP against the euro still exists.

Going further, we elaborate on factors driving the CCB. The presence of nonlinearities requires the application of methods from deep learning. We train and test a multilayer perceptron (MLP) and show how it adds value. MLPs have been used for such purposes for decades (e.g., Volmer and Lehrbass 1997). To avoid overfitting, parsimonious modeling and a large testing period are applied. To our knowledge, we are the first to apply deep learning methods to the topic.

Thus, "we synthesize the empirical … pricing literature with the field of machine learning. Relative to traditional empirical methods in asset pricing, machine learning accommodates … richer specifications of functional form" (Gu, Kelly, and Xiu 2017).

Not all nonzero levels of the CCB imply an arbitrage opportunity: "It has been long acknowledged that deviations from CIP may be present while satisfying the absence of arbitrage opportunities, as long as they are contained within the neutral band" (Hernandez 2019). Nonetheless, this neutral band is difficult to quantify. A key factor is the synchronicity of the data: "It is important to have data on the appropriate exchange rates and interest rates recorded at the same instant in time at which a trader could have dealt" (Taylor 1987).

Global banks are rather well positioned to exploit arbitrage opportunities. "They are present in funding markets in multiple currencies, face constant funding/liquidity needs, and can flexibly choose the cost-optimal funding option. Other important market participants, for example, hedge funds, are less likely to engage as major arbitrageurs" owing to worse funding conditions (Rime, Schrimpf, and Syrstad 2016). However, one has to differentiate among banks because empirically only "a narrow set of global banks has enjoyed risk-less CIP arbitrage opportunities" and "there is an important asymmetry between top-rated and lower-rated banks" (Rime, Schrimpf, and Syrstad 2016). Furthermore, more differentiation is necessary on a lower level—for example, whether "a trader has a natural long position (i.e. an endowment) in a currency" (Taylor 1987). Finally, there is execution risk involved in the exploitation of arbitrage opportunities, which might also differ among the potential arbitrageurs.

All in all, this means that we cannot quantify trading gains and related measures of risk and return. Nevertheless, our modeling is guided by a commercial perspective because we focus on explaining and forecasting the daily levels of the CCB.

Our selection of explanatory variables is inspired by the literature. Jauregui and Natraj (2017) showed that monetary policy events matter. Du, Tepper, and Verdelhan (2018) included regulatory reporting dates and funding, transaction, and balance sheet costs. Dao, McGroarty, and Urquhart (2019) considered the Brexit vote itself and found significant impacts on the currency market.

The benefits of our contribution are manifold. Arbitrage desks can prepare for days when there are large arbitrage gains. Corporates can punctually adapt their procurement of currencies that are about to become scarce. Researchers may deepen their understanding of market mechanics. In addition, we hope to contribute to the vision of White: "Application of neural networks to new and existing datasets holds the potential for fundamental advances in empirical understanding across a broad spectrum of sciences. To realize these advances, statistics and neural network modeling must work together, hand in hand" (White 1992).

The article is organized as follows. The following section introduces key concepts and the subsequent one describes the data. Thereafter, a section is devoted to modeling; it introduces the key metrics and contains the econometric modeling and the deep learning approach. A final section concludes.

CIP CONDITION AND CROSS-CURRENCY BASIS

In this section, we review the CIP condition and define the CCB as the deviation from the CIP condition as done by Du, Tepper, and Verdelhan (2018). Let $y_{t,t+n}^{EUR}$ and $y_{t,t+n}^{GBP}$ denote the continuously compounded $1/n$-year risk-free interest rates quoted at date t in euro and GBP, respectively. The spot exchange rate S_t is expressed in units of GBP per euro; an increase in S_t thus denotes a depreciation of the GBP and an appreciation of the euro. Likewise, $F_{t,t+n}$ denotes the n-year outright forward exchange rate in the same units at time t. The CIP condition states that the forward rate should satisfy

$$e^{ny_{t,t+n}^{EUR}} = e^{ny_{t,t+n}^{GBP}} \frac{S_t}{F_{t,t+n}} \tag{1}$$

Henceforth, we use the natural logarithms of the two exchange rates and use lowercase letters instead of capital ones. Thus, the CCB $x_{t,t+n}$ is quantified as

$$x_{t,t+n} = y_{t,t+n}^{EUR} - \left\{ y_{t,t+n}^{GBP} - \frac{1}{n}(f_{t,t+n} - s_t) \right\} \tag{2}$$

When the CIP condition is fulfilled, the level x of the CCB amounts to zero. In the case of a positive basis, the arbitrage strategy is as follows: Start with funding in the synthetic euro risk-free rate by taking a loan in GBP, exchange it in the spot market for euros, and buy GBP at the forward exchange rate to hedge the foreign exchange risk at loan maturity. Then come full circle by investing in the direct euro risk-free rate. In total, this strategy would yield an annualized risk-free profit equal to x percentage of the trade notional.

EMPIRICAL DATA AND POTENTIAL DRIVING FACTORS

Du, Tepper, and Verdelhan (2018) included the nominal interest rate differential, which is the difference between foreign and domestic interest rates, into the set of regressors. In light of Equation 2, it is obvious that this might cause a problem of endogeneity.[1] Therefore, we do not consider the nominal interest rate differential as a potential factor in the sequel.

Because we model the impact of the Brexit, we have to go back to the very beginning. On May 27, 2015 the European Union Referendum Bill was unveiled in the Queen's Speech. This was the UK legislation required to allow the referendum to take place in June 2016. Hence, our data start on May 27, 2015 and reach to the end of 2019. We obtain daily spot closing exchange rates[2] and forward points[3] as well as interbank interest rates from Bloomberg. The Euro Interbank Offered Rate (EURIBOR)[4] and the London Interbank Offered Rate (LIBOR)[5] are used as a proxy for the European and British funding rates. All interest rates are continuously compounded. With these figures, we calculate the CCB.

The first potential driving factor of the CCB is credit risk—more specifically, the difference between foreign and domestic credit risk. The credit risk is approximated as the interest rates used for funding minus the risk-free interest rate. The interbank interest rates EURIBOR and LIBOR are determined daily via a panel. They are used for funding in the EU and United Kingdom. To obtain the credit risk, we deduct the risk-free interest rates as given by the overnight interbank interest rates EONIA[6] for the EU and SONIA[7] for the United Kingdom. UK credit risk minus EU credit risk constitutes the factor CR.

The second factor is transaction costs. The transaction costs are approximated by the absolute value of the bid–ask spread of the CCB. We call this factor *TC*.

The third factor is the balance sheet costs of the banks. We deduct the interest on excess reserves (IOER) from one-week EURIBOR and call this factor *BC*. We use the deposit rate[8] of the European Central Bank as a proxy for IOER.

Furthermore, we analyze the impact of certain events. First, we examine the effect of regulatory reporting dates. Hence, we use an indicator variable, which assumes a value of one on business days before a regulatory reporting date and zero otherwise. We call this factor *Qend*.

Second, we examine the impact of monetary policy events by looking for changes in base rates of the European Central Bank and the Bank of England. The indicator variable assumes a value of one on dates of changes in base rates as well as one day before and zero otherwise. We do so because the announcement of changes often takes place on the day before the change. We call this factor *MP*. Lastly, we track Brexit-related politics via an indicator variable, which assumes a value of one on dates on which significant information relating to the Brexit was published and zero otherwise. We call this factor *BN*. Note that commercial reasoning lets us expect a positive coefficient for all factors.

MODEL BUILDING

Data Split and Key Metric

In contrast to Du, Tepper, and Verdelhan (2018), we split the sample. To be able to check for overfitting, we only use data from 2015 to the end of 2017 for training/

[1] We thank Jürgen Kähler, PhD, professor of economics at the University Erlangen-Nuremberg, for this helpful comment. He also hinted at the articles by Taylor (1987) and Meese and Rogoff (1983), which we have taken into consideration.
[2] Bloomberg-ticker: EURGBP BGN Curncy.
[3] Bloomberg-ticker: EURGBP1W BGN Curncy.
[4] Bloomberg-ticker: EUR001W Index.
[5] Bloomberg-ticker: BP0001W Index.
[6] Bloomberg-ticker: EUSWE1Z BGN Curncy.
[7] Bloomberg-ticker: BPSWS1Z BGN Curncy.
[8] Bloomberg-ticker: EUORDEPO Index.

EXHIBIT 1

Diagnostics for Full Regression Model

Check	H0	Statistics	*P*-Value
RESET	No specification error	67.869	Close to zero
Durbin–Watson	No autocorrelation of first order	1.2578	Close to zero
Breusch–Pagan	Homoskedasticity	269.01	Close to zero
Variance Inflation Factors		All below 1.32	

EXHIBIT 2

Estimation Results for Full Regression Model

Factor	Estimate (SE)	*P*-Value
Intercept	–2.954529 (4.445998)	0.5065858
CR	0.229198 (0.604597)	0.7047445
TC	**0.650013 (0.191398)**	**0.0007257**
BC	**0.936804 (0.508387)**	**0.0658346**
Qend	–2.366859 (3.354054)	0.4806491
MP	0.014916 (2.534414)	0.9953060
BN	**23.356256 (10.475221)**	**0.0261152**

estimation. The models are tested in the years 2018 and 2019. Thus, our in-sample set consists of 648 days, and the out-of-sample set covers 501 days. Because we focus on explaining and forecasting daily levels of the CCB, the squared correlation of forecasted and actual levels is our key metric (or R^2) in judging on model performance. Throughout, we use the open-source statistics software R via RStudio.[9]

Econometric Approach

Like Du, Tepper, and Verdelhan (2018), we first use a multiple linear regression. The usual tests for stationarity show that all time series under consideration are stationary.[10] Therefore, we estimate a multiple linear regression[11] and apply the usual diagnostics as shown in Exhibit 1.

The model achieves an R^2 of 0.4011 but needs further consideration. The linear specification seems to be inadequate and/or important factors are missing. This will be addressed in the sequel in which we use an MLP, in which estimation and specification go hand in hand. Missing factors are the fate of many models because they are only approximations of reality. The last line shows that at least there are no redundant factors. There is autocorrelation and heteroskedasticity in the residuals. This requires the usage of Newey–West standard errors (SE), which are shown in Exhibit 2 in parentheses.[12] The estimators for the coefficients are consistent, and the test statistics "estimate divided by SE" are normally distributed (Verbeek 2012).

For the factor Qend, the estimated coefficient is negative. Because we cannot reject H0 that the true coefficient is zero, the sign does not cause a problem. The factor CR is insignificant, too. This might be because most arbitrage deals happen between banks with similar creditworthiness. Only the factors TC and BN are highly significant; BC is significant at a level of 10% only. As we are heading toward an MLP, it would be too harsh to focus only on the two highly significant factors in what follows. It could be that the less significant factor BC adds value beyond linearity. Hence, we keep it. We estimate a linear model with these three factors. We keep the intercept. Thus, the R^2 stays between zero and one, and the mean of the residuals is close to zero. The model still

[9]We use RStudio and the R packages: car, tseries, sandwich, zoo, lmtest, and mxnet.

[10]With respect to a different dataset, Du, Tepper, and Verdelhan (2018) are ambiguous on the order of integration of the CCB. They switched between using the level of the CCB versus its first difference (e.g., Equations 13–14 versus Equations 9 and 17).

[11]Verbeek (2012) gives a good introduction. We use the least squares method, which has an analytic solution, for estimation.

[12]With this amount of data, the Newey–West correction appears applicable. For details, see Verbeek (2012).

EXHIBIT 3
Diagnostics for Focussed Regression Model

Check	H0	Statistics	*P*-Value
RESET	No specification error	62.821	Close to zero
Durbin–Watson	No autocorrelation of first order	1.2537	Close to zero
Breusch–Pagan	Homoskedasticity	269.43	Close to zero
Variance Inflation Factors		All below 1.07	

EXHIBIT 4
Estimation Results for Focussed Regression Model

Factor	Estimate (SE)	*P*-Value
Intercept	–2.95453	0.399914
	(3.50755)	
TC	**0.65001**	**0.000195**
	(0.17347)	
BC	**0.93680**	**0.028247**
	(0.42605)	
BN	**23.35626**	**0.025682**
	(10.44463)	

EXHIBIT 5
Stationarity Tests for Residuals of Regression Model

Test	Statistics	*P*-Value
Augmented Dickey–Fuller	–6.7493	Smaller than 1%
Phillips–Perron Unit Root	–389.22	Smaller than 1%
KPSS	0.32555	Larger than 10%

EXHIBIT 6
Performance of Two Benchmark Models

Consideration	R^2 Random Walk	R^2 Three-Factor Linear Regression
In-Sample Static	0.3389	0.3997
Out-of-Sample Static	0.1622	0.1426
Out-of-Sample Day-Ahead Running	0.1622	0.1908

NOTES: The random walk model shows a sharp drop in its R^2 out of sample, which shows that the out-of-sample period is a special one.[14] This supposition is confirmed by an even larger drop in R^2 for the linear regression model.[15]

achieves an R^2 of 0.3997 but also needs diagnostics, which paint a similar picture as before (Exhibit 3).

The RESET test still rejects the H0.[13] Again, we report the estimates and their Newey–West SE (Exhibit 4).

It turns out that now all three factors are significant at a level of 3%. The resulting residuals are stationary, as three tests support (Exhibit 5).

For practical purposes, stationarity of the forecasting errors is an important feature. In our case, it means that they fluctuate around a mean of zero with a tendency to return to their mean.

We consider this linear regression model as our first benchmark model. "This benchmark has a number of attractive features. It is parsimonious and simple, and comparing against this benchmark is conservative because it is highly selected (the characteristics it includes are routinely demonstrated to be among the most robust return predictors)" (Gu, Kelly, and Xiu 2017). There is a second benchmark model, which treats today's CCB as the best forecast for tomorrow's. It is called a *random walk model*.[16]

The performance of the two benchmark models is judged threefold: in sample, out of sample, and via a running estimation and day-ahead forecasting (Exhibit 6). In the latter, a model is estimated each day using data up to $t - 1$. A constant window of the last 64 days[17] is used for estimation to avoid using information that is too outdated. For the linear regression model, the estimated coefficients are used to forecast

[13] The RESET test yields a *p*-value of close to zero. This can be read as suggesting that the functional (linear) form might be insufficient. This could be treated by inclusion of quadratic, cubic, and higher terms of the factors. Plotting the residuals and looking at their squared autocorrelations points toward higher-order serial dependencies. This could be handled by modeling a GARCH error term. Instead of doing either, we directly apply a more flexible deep learning approach in the sequel.

[14] We calculate R^2 as the squared correlation of the CCB and its forecast.

[15] To further explore these levels of R^2, we estimate the three-factor linear regression model directly on the test set. An R^2 of 0.1721 emerges.

[16] For example, in an important paper on foreign exchange models: "The random walk model, which uses the current spot rate as a predictor of all future spot rates" (Meese and Rogoff 1983).

[17] Choosing a window size of 64 (i.e., a bit more than three months of trading days) conforms to the way batch sizes are chosen. Both figures correspond. The batch size must not be larger than the window.

the CCB at time *t*, using the factors as of *t*. Hence, for estimation, no information beyond *t* – 1 is used.

Deep Learning Approach

The "universal approximation theorem states that a feedforward network with a linear output layer and at least one hidden layer … can approximate … any continuous function" (Goodfellow, Bengio, and Courville 2016) and that "with two layers, even discontinuous functions can be represented" (Russell and Norvig 2016). Therefore, we use an MLP with two hidden layers. To avoid overfitting, we choose a slim architecture. A rule of thumb is to set the number of hidden neurons at two-thirds of the number of factors + 1. With three factors in place, this rule leads to using three hidden neurons.[18]

We choose the number of neurons in each layer according to the geometric pyramid rule: "for many practical neural networks, the number of neurons follows a pyramid shape, with the number decreasing from the input toward the output" (Masters 1993). Hence, we place two hidden neurons on the first hidden layer (closest to the input layer) and one in the second hidden layer, which is followed by a linear output neuron.

As in the previously mentioned regression, we seek to minimize the Euclidean distance between forecasted and realized daily levels of the CCB in sample. In the regression, we find the model parameters that minimize this loss function analytically, whereas here we search for such weights[19] numerically. We call this search process *training*. Note that estimation (training) and specification happen at the same time.

The in-sample (out-of-sample) dataset is now called training data (test set). As is usual for the training of an MLP, we apply min–max normalization, which is merely a linear transformation of the original data. A minibatch stochastic method is applied (Goodfellow, Bengio, and Courville 2016). Because "small batches can offer a regularizing effect," we use a small batch size of 16 (Goodfellow, Bengio, and Courville 2016). To select a specific learning algorithm for backpropagation[20] a maximum of 100 training epochs is used. This is as done by Lim, Zohren, and Roberts (2019)

LEARNING ALGORITHM

Because we use the R package MXNet, we start our training with settings as in the MXNet Documentation, Release 0.0.8. In this document, a simple three-layer MLP is presented with relu activations, which effectively is the same as our two-hidden-layer feedforward network. For training a basic stochastic gradient descent (SGD[21]) algorithm with a learning rate of 0.1, a momentum of 0.9 and a weight decay at 0.00001 are applied. Initialization of the weights is done using drawings from a standard uniform distribution centered at zero with a radius of 0.01. We fix the seed for the random number generator to make the results reproducible.

Using relu activations, there is no progress in learning. Therefore, we switch to hyperbolic tangent activation, giving us stronger gradients.[22] The resulting architecture

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[18] As concerns the richness of the architecture, we hope to "stick with the minimum number necessary to solve the problem" (Masters 1993).

[19] The model parameters of a neural network are called *weights*. See the appendix for more detail.

[20] We assume the readers' familiarity with backpropagation. A good introduction is given by Goodfellow, Bengio, and Courville (2016).

[21] A description of SGD is given by Goodfellow, Bengio, and Courville (2016).

[22] The gradient is the vector of first partial derivatives of the loss function with respect to the model parameters (i.e., the weights).

is shown in the appendix. Even with the change of activations toward tanh, the results are still poor. In sample, the R^2 is only 0.008, which is worse than the linear regression. In theory, the MLP should encompass a linear function easily, but it is more promising to replace this default algorithm.

Some algorithms[23] with adaptive learning rates are available in MXNet. Using their default hyperparameters allows a quick comparison. Permanent adaptation is not harmful because the essential information of the gradient is the direction of the weight update and not its amount per se. It also is important that the adaptation eventually approach zero for backpropagation to settle down.[24]

The best performance in sample is achieved by AdaGrad, which also "enjoys some desirable theoretical properties" (Goodfellow, Bengio, and Courville 2016). Therefore, we change the training algorithm from SGD to AdaGrad (Duchi, Hazan, and Singer 2011). This algorithm "adapts the learning rate of all model parameters by scaling them inversely proportional to the square root of the sum of all the historical squared values of the gradient" (Goodfellow, Bengio, and Courville 2016). There is, however, the warning that "for training deep neural network models, the accumulation of squared gradients from the beginning of training can result in a premature and excessive decrease in the effective learning rate" (Goodfellow, Bengio, and Courville 2016). In light of our slim architecture, we do not have to worry. Hence, we can give more detail about AdaGrad and our choice of hyperparameters.

Hyperparameter Tuning

The global learning rate ε plays a crucial role.[25] Therefore, we begin by searching for the best parameter value in the set (0.01, 0.05, 0.1, 0.2, …, 0.9). The default value for the learning rate in MXNet is 0.05. However, a learning rate of $\varepsilon = 0.7$ achieves the highest R^2 in sample. There is also a small constant δ in the algorithm, which we leave at its default value of 10^{-7} (Goodfellow, Bengio, and Courville 2016).

In the sequel, capital letters signify vectors. We denote the weight vector by W, which is changed during training as follows. First, keep track of the gradient's G history by accumulating the squared values in a vector R, which is initialized with a vector of zeros. Note that $A \circ B$ denotes the elementwise (Hadamard) product of A and B. Thus, we can specify the process of accumulation:

$$R \leftarrow R + G \circ G \tag{3}$$

Second, divide and square root elementwise to update the corresponding entry in the weight vector W by changing it by the amount ΔW as follows:

$$\Delta W = -\frac{\varepsilon}{\delta + \sqrt{R}} \circ G \tag{4}$$

The advantage of AdaGrad is that it assigns "frequently occurring features very low learning rates and infrequent features high learning rates, where the intuition is that each time an infrequent feature is seen, the learner should 'take notice'" (Duchi,

[23] The set consists of SGD, AdaGrad, and RMSProp. A good description is given by Goodfellow, Bengio, and Courville (2016). AdaGrad stands for adaptive gradient algorithm. In addition, there is a modification of AdaGrad called AdaDelta, which is described by Zeiler (2012).

[24] For more details, see White (1992).

[25] The learning rate is that fraction of the gradient that is used for adaptation of the weights during training to minimize the loss function. "Neural network researchers have long realized that the learning rate is reliably one of the most difficult to set hyperparameters because it significantly affects model performance" (Goodfellow, Bengio, and Courville 2016).

EXHIBIT 7

Stationarity Tests for Residuals of MLP

Test	Statistics	*P*-Value
Augmented Dickey–Fuller	–6.9567	Smaller than 1%
Phillips–Perron Unit Root	–381.57	Smaller than 1%
KPSS	0.37218	0.08914

EXHIBIT 8

Performance Comparison Regression Model vs. MLP

Consideration	R^2 Three-Factor Linear Regression	R^2 MLP
In-Sample Static	0.3997	0.4646
Out-of-Sample Static	0.1426	0.0760
Day-Ahead Running	0.1908	0.3140

Hazan, and Singer 2011). Especially for purposes of trading and arbitrage, the infrequently occurring features (i.e., datasets) matter. They are often related to big market moves and offer great commercial potential.

We have estimated the parameters of the linear regression without the help of cross-validation. Furthermore, the regression model has been estimated using only in-sample data. Therefore, we do not apply a validation set during training and do not use early stopping. Instead, we monitor the loss function during training to judge the necessary number of epochs and try out sensible figures for the maximum number of epochs.

It is common to consider epochs up to 250 (Goodfellow, Bengio, and Courville 2016). Increasing the maximum number of training epochs from 100 to 250 yields an increase in the R^2 in sample from 0.4633 to 0.4639. Indeed, monitoring the evolution of the error shows that there is steady progress in learning. This motivates consideration of other maxima, such as 500, 1,000, 1,500, 2,000, and 2,500. Again, we get higher R^2 values of 0.4642, 0.4644, 0.4646, 0.4647, and 0.4648. In light of the decreasing up-moves in R^2, we set the maximum number of epochs to 1,500 to provide a level playing field for comparison of the two approaches. Thus, we allow for sufficient time for the numerical search. We do not fear overlearning in light of the chosen slim architecture. Instead, we have some confidence in the thus inherent generalization capacity of our model. The weights of the MLP are shown in the appendix.

The mean of the residuals is close to zero.[26] Two of three tests indicate that the residuals—resulting from forecasting with the MLP—are stationary (Exhibit 7).

Hence, again we encounter a relationship, but this time it is nonlinear.

PRACTICAL PERFORMANCE

Because the MLP has 13 model parameters instead of only four, as in the three-factor linear regression, a significant increase in R^2 can be expected in sample. Exhibit 8 shows that the MLP performs better than the linear regression model in sample but drops below out of sample. The drop in R^2 is larger than that for the linear regression model.[27]

A real-life application would not leave the models untrained for two years. Hence, a running day-ahead forecast is the most relevant one from a commercial perspective. Because we use a small architecture with only 13 weights, we can still afford the estimation window of 64 days[28] for training. Here, the functional flexibility has paid off. The R^2 is significantly higher. This makes the MLP a worthwhile approach.

[26] It amounts to –0.0007531197. Hence, there does not seem to be a systematic error.

[27] Using all factors/regressors would enlarge this effect. The linear regression would drop from an R^2 of still 0.4011 to 0.1391, but the MLP would decrease from 0.5055 to merely 0.0314. This shows that focusing the set of regressors based on econometrical testing is helpful.

[28] This is not unusual. "If there are only two inputs, a dozen or so training examples will probably cover every error pattern, at least for all practical purposes" (Masters 1993). Note that we choose window and batch size as powers of 2. Thus, interpretation is easier. Here, every fourth dataset is used in the approximation of the gradient.

EXHIBIT 9

Elasticities of Factors in Regression Model vs. MLP

Factor	Elasticity in Linear Model	Elasticity in MLP
TC	0.8301	0.0732
BC	0.0091	0.0025

But can we make sense of the results? To interpret both models, we calculate sensitivities in sample. We exclude the dummies and the intercept from this consideration. We increase each factor[29] by 1% and calculate the percentage increase in the forecasted CCB. Division of the latter by the former produces the elasticities shown in Exhibit 9.

The models agree with regard to the signs but disagree on the relative importance. For the two highly significant factors TC and BC, the MLP is less sensitive than the linear model.[30]

CONCLUSION

The market for GBP versus euro seems to offer daily arbitrage opportunities. The CCB is driven by its transaction costs, the balance sheet costs of the banks, and news concerning Brexit. Three models have been compared to understand and forecast the CCB. Using them, market participants can reduce the degree to which they are taken by surprise. Arbitrage desks can prepare days in advance when there might be large upcoming arbitrage opportunities. Corporates can procure currencies that are about to become scarce in advance. Moving from a linear regression model toward an MLP increases the R^2 for in sample and running day-ahead forecasts considerably.

APPENDIX

The architecture of the MLP is as follows: The input vector X has one row and three columns, as in the linear regression model. The weight vector in the kth hidden layer is W^k a column with three rows for $k = 1$ and two rows for $k = 2$. The corresponding bias weights are b^k.

The hidden neurons in the kth hidden layer are denoted by h_i^k with i running from 1 to 2 in the first layer and being equal to 1 in the second layer. The activation functions obey

$$h_i^1 = \tanh(XW^1 + b_i^1)$$

$$h_1^2 = \tanh\left(\sum_1^4 W_i^2 h_i^1 + b^2\right)$$

Using the notation as in the MXNet Documentation, Release 0.0.8, the first (second) equation corresponds to fc1 (fc2) in the sequel. At last, there is the linear output neuron corresponding to fc3. The weights of the MLP result from training in sample are shown below. We denote the learned weights all together by $\hat{W}$ and supply their numerical values as follows:

```
$fc1_weight
             [,1]        [,2]
[1,] 6.123996e-14 2.97964644
```

[29] The other factors are kept at their mean level. The dummies are fixed at zero. We use a three-factor regression based on min–max transformed data to make figures comparable.

[30] We only note that interaction effects can be investigated if "we simultaneously vary values of a pair of" factors (Gu, Kelly, and Xiu 2017).

```
[2,] -4.806135e-12 0.03480449
[3,] -1.312563e-12 0.59536171

$fc1_bias
[1] 8.067019e-13 -1.353642e+00

$fc2_weight
             [,1]
[1,] 1.648870e-14
[2,] 1.121383e+00

$fc2_bias
[1] 0.01990237

$fc3_weight
          [,1]
[1,] 0.3444811

$fc3_bias
[1] 0.3876643
```

The model has two hidden neurons in the first layer, into which three factors are fed. Together with the bias weight, the number of parameters is eight. In the second hidden layer, there is only one neuron, receiving two inputs from the preceding hidden layer. This adds three parameters. The output layer is linear (i.e., two more parameters). In total, the MLP has 13 parameters.

Concerning the individual factors, it is possible to carry out hypothesis testing as for the linear regression model. Like the estimates in the linear regression model, the reported weights are random variables. "Formal statistical inference … is possible whenever the limiting distribution of $\hat{W}$ is known" (White 1992). For this to be the case, however, we need two more necessary conditions to be fulfilled: (1) There are no redundant inputs and (2) there are no irrelevant hidden units (White 1992). The latter is harder to show, and arguing both would take us beyond the scope of this article.

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